

Control Statements

Control Statements are the statements which can change the flow or order of execution of the instrument.

Indentation

- ```
if a > b:
 print("Hello")
else:
 print("Hi")
```
- ```
marks = int(input("Enter marks:"))  
if marks > 90:  
    print("Grade A1")  
elif marks > 80:  
    print("Grade A2")  
elif marks > 70:  
    print("Grade B1")  
else:  
    print("Other grade")
```

Nested IF

- ```
score = int(input("Enter score"))
if score > 90:
 if score >= 100:
 print("Century")
 else:
 print("Well played")
else:
 print("Better play next time")
```

### IF/ELSE

- ```
num = 20  
if num % 2 == 0:  
    print("Even")  
else:  
    print("Odd")
```

Multi-Variable Assignment

```
a, b, c = 1, 2, 3  
print(a) = 1  
print(b) = 2  
print(c) = 3
```


$x, y, z = 10, 5, 2$

if $x > 5$:

$x = x \gg y // \sim z$

if $x > 12$:

print(x)

else:

print(z)

else:

print(y)

Loops

When we repeatedly want to perform
then we use loops.

for

while

when you know how
many times loop will
be executed

when you do not know
how many times loop will
be executed.

• routine = ["wakeup", "breakfast", "study", "revision", "sleep"]

for item in routine:

print(item)

O/P $\Rightarrow$

wakeup
breakfast
study
revision
sleep

• lst1 = [10, 20, 30, 40]

for item in lst1:

print(item)

O/P $\Rightarrow$
10
20
30
40

• for item in range(len(lst1)):

print(lst1[item])

$\left\{ \begin{array}{l} 10 \\ 20 \\ 30 \\ 40 \end{array} \right\}$ O/P

Control Stmtz.

$\Rightarrow$ If, Else, ... $\rightarrow$ Else in loop
 $\Rightarrow$ Loops $\rightarrow$ small case
 $\Rightarrow$ zip, break, cont.

• `lst1 = [10, 20, 30, 40]`
 for `i, v` in `enumerate(lst1)`:
 `print(i, v)`

O/p ⇒

0	10
1	20
2	30
3	40

Nested loop

players = [["Kohli", "Rohit"],
 ["Ponting", "Clarke"],
 ["Bawuma", "Miller"]]

for `a` in `players`:

 for `b` in `a`:
 `print(b)`

O/p ⇒ Kohli
 Rohit
 Ponting
 Clarke
 Bawuma
 Miller

★ zip()

• parties = ["Apple", "Banana", "Guava"]

seats = [100, 200, 50]

for `a, b` in `zip(parties, seats)`:

`print(a, b)`

O/p ⇒ Apple 100
 Banana 200
 Guava 50

Note: If there are more indexes in another list, then it can give output where there is only proper index length. Eg: `lst1 = 3, lst2 = 3` O/p ⇒ 3
 `lst1 = 4, lst2 = 3` O/p ⇒ 3 only paired

• `l1 = [10, 20, 30, 40]`

`l2 = (11, 21, 31)`

`l3 = {100, 200, 300}`

`l4 = "Virat"`

for `a, b, c, d` in `zip(l1, l2, l3, l4)`:

`print(a, b, c, d)`

O/p ⇒

10	11	100	"V"
20	21	200	"i"
30	31	300	"r"

It can be

100	200	300
200	300	200
300	100	100

• fruits = ["apple", "banana", "orange"]

i = 0
n = len(fruits) = 3

while i < n:

print(fruits[i])

i = i + 1

O/P => apple

banana

orange

* Break:

Whenever the break statement get executed, stop the iteration & just immediately come out of the loop.

• lst1 = [30, 20, 40, 10, 60, 70, 80]

total = 0

for item in lst1:

total += item

if total > 100: 160 > 100

break

print(total)

item = 30, 20, 40, 10, 60

O/P => { 30, 50, 90, 100 }

* Continue:

Continue statement is used to stop the current iteration and jump to the next iteration. It does not terminate the loop.

• lst1 = [5, -3, 7, 6, -3, -4, 9, 11, -2]

total = 0

for item in lst1:

if item < 0: -3, -3, -4, -2

continue

total += item

print(total)

O/P => { 5, 12, 18, 27, 38 }

• ll = [7, 5, 10, 12, 15]

for i in ll:

if i % 5 == 0: 5, 10, 15

continue

print(i)

O/P => { 7, 12 }

• Pass: (Empty stmt.) Might add something in future

Pass is like a placeholder where you can write some code in future.

```
l1 = [10, 20, 30, 40, 50]
```

```
for a in l1:
```

```
    if a > 30:
```

```
        pass
```

```
    print(a)
```

O/p ⇒

```
10
20
30
40
50
```

• Else in loop:

```
l1 = [10, 20, 30, 40]
```

```
for items in l1:
```

```
    if item > 40:
```

```
        break
```

```
    print(item)
```

```
else:
```

```
    print("Else Block")
```

Terminated by break

item > 30 → O/p ⇒

```
10
20
30
```

(No else block)

O/p ⇒

```
10
20
30
40
Else Block
```

Note := Else Block will be executed after finishing loop only when break statement has not got executed

• Match-Case:

match case is similar to switch in other programming languages which is used to match pattern.

⇒ match <expression>:

```
case <pattern>:
```

```
    //code
```

```
case <pattern>:
```

```
    //code
```

```
case <pattern>:
```

```
    //code
```

```
case --: } default
    //code.
```

```
day = "monday"
```

```
match day:
```

```
    case "monday":
```

```
        print("First day")
```

```
    case "Sunday":
```

```
        print("Last day")
```

```
    case --:
```

```
        print("Other day")
```

O/p ⇒ First day.


```

• fruit = "apple"
match fruit:
    case "apple" | "orange":
        print("Fruits")
    case _:
        print("Vegetables")

```

O/P $\Rightarrow$ Fruits.

```

• l1 = [1, 2, 3]
match l1:
    case [1, 2, 3]:
        print("Hello")
    case [1, *a]:
        print(a)

```

O/P $\Rightarrow$ "Hello".

Practice Questions.

Q1 $i = 1$
 while $i < 10$:
 print(i)
 $i = i // 2$

1
0
0
0

$\Rightarrow 1 // 2 = 0.5$
 $= 0$

Q3 count = 0
 for i in range(4):
 for j in range(4):
 if $i = 2$ and $j > 1$:
 continue
 count += 1
 print(count)

Q2

count = 0
 for i in range(3):
 for j in range(3):
 if $i + j < 3$:
 count += 1

O/P = 6

print(count)

i = 0: j = 0 C = 1
 j = 1 C = 2
 j = 2 C = 3
 i = 1: j = 0 C = 4
 j = 1 C = 5
 j = 2 C = 6
 i = 2: j = 0 C = 7
 j = 1 C = 8
 j = 2 C = 9

i = 0: j = 0 C = 1
 j = 1 C = 2
 j = 2 C = 3
 j = 3 C = 4
 i = 1: j = 0 C = 5
 j = 1 C = 6
 j = 2 C = 7
 j = 3 C = 8
 i = 2: j = 0 C = 9
 j = 1 C = 10
 j = 2 X
 j = 3 X
 i = 3: j = 0 C = 11
 j = 1 C = 12
 j = 2 C = 13
 j = 3 C = 14

point = (1, 2)

match point:

case (0, 0):
 print("GATE")

case (1, y):
 print("2027")

case (x, 2):
 print("Hello")

case (x, y):
 print(x, y)

O/P $\rightarrow$ 2027

Q=4

K=0
for i in range(5):
 for j in range(i,5):
 K += i+j
 print(K)

i=0
j=0 K=0
j=1 K=1
j=2 K=3
j=3 K=6
j=4 K=10

i=1
j=1 K=12
j=2 K=15
j=3 K=19
j=4 K=24

i=2
j=2 K=28
j=3 K=33
j=4 K=39

i=3
j=3 K=45
j=4 K=52

O/P → 60

Q=5

for i in range(3):
 for j in range(2):
 if i==j:
 break
 else:
 print(i)

i=0
j=0 break

i=1
j=0
j=1 break

i=2
j=0
j=1

O/P = 2

Q=6

count=0
for i in range(1,4):
 for i in range(1, i+1):
 count+=1
 print(count)

i=1
j=1 c=1
j=2 c=2
j=3 c=3

Q=7

count=1
for i in range(1,10,2):
 count=count+1
 while count<=20:
 count=count+2
 print(count)

i=3
c=2 → c=4, 6, 8, 10, 12, 14, 16, 18, 20
i=5
c=24 → c=26
i=7
c=25
i=9
c=26

26

